K.R. MANGALAM UNIVERSITY

Database Management Lab File
(ENCS205)

Bachelor of Technology
in
Computer Science and Engineering

Submitted by
Krishna Vohra (2401010066)

B. Tech CSE (Core) – 4 Semester

Submitted To:
MS. Mansi Kajal

School of Engineering & Technology

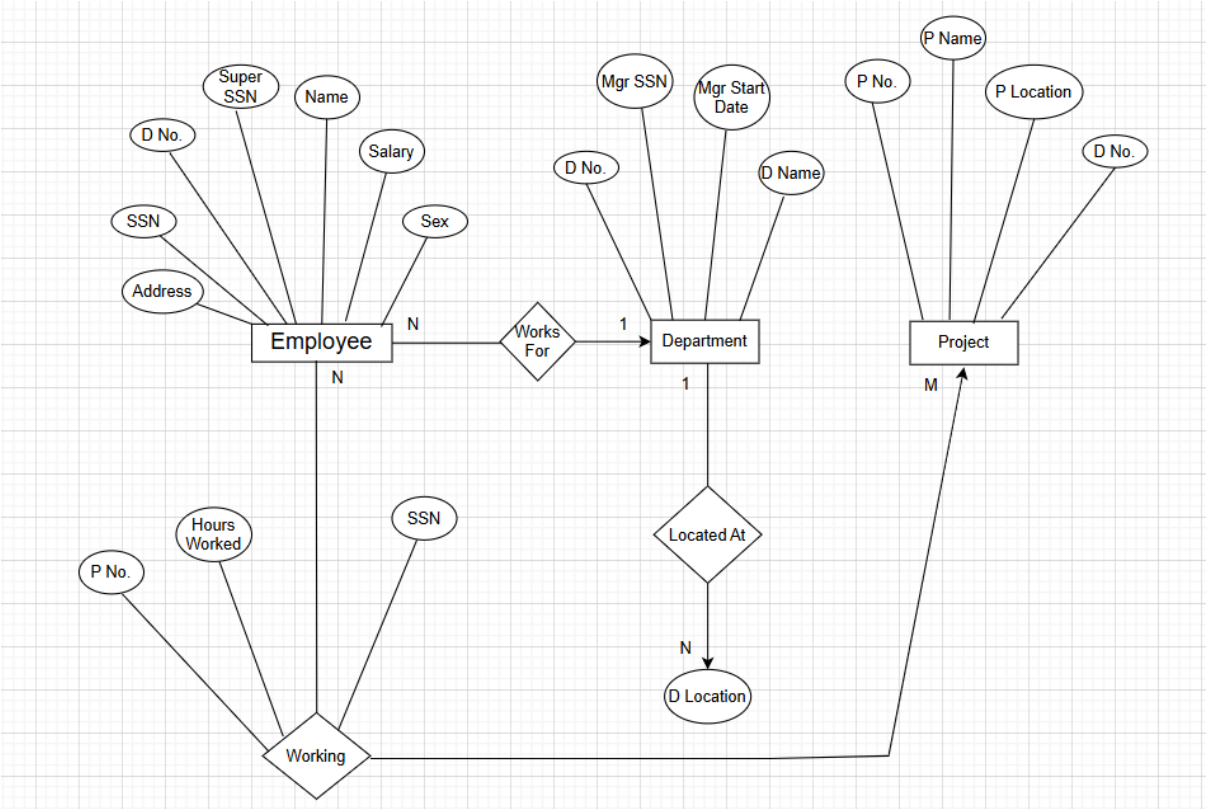
K. R. MANGALAM UNIVERSITY Sohna, Haryana
122103, India Jan-2026

Experiment 1

1. **Consider following databases and draw ER diagram and convert entities and relationships to relation table for a given scenario:**

COMPANY DATABASE:

EMPLOYEE (SSN, Name, Address, Sex, Salary, Super SSN, D No.) DEPARTMENT (D No., D Name, Mgr SSN, Mgr Start Date) DLOCATION (D Location)
PROJECT (P No., P Name, P Location, D No.)
WORKING (SSN, P No., Hours Worked)



Explanation of ER Diagram – Company Database :

- This ER diagram represents a company database that stores information about employees, departments, locations, and projects.

- The Employee entity stores details such as SSN, name, address, sex, salary, supervisor SSN, and department number. Each employee works for a department. This relationship is shown using WORKS FOR. Many employees can work for one department, but each employee works for only one department.
- The Department entity stores department information such as

department number, department name, manager SSN, and manager start date. A department can have multiple locations. This is shown using the LOCATED AT relationship between Department and D Location.

- The Project entity stores information about projects such as project number, project name, project location, and department number. Each project is controlled by one department, and one department can control many projects.

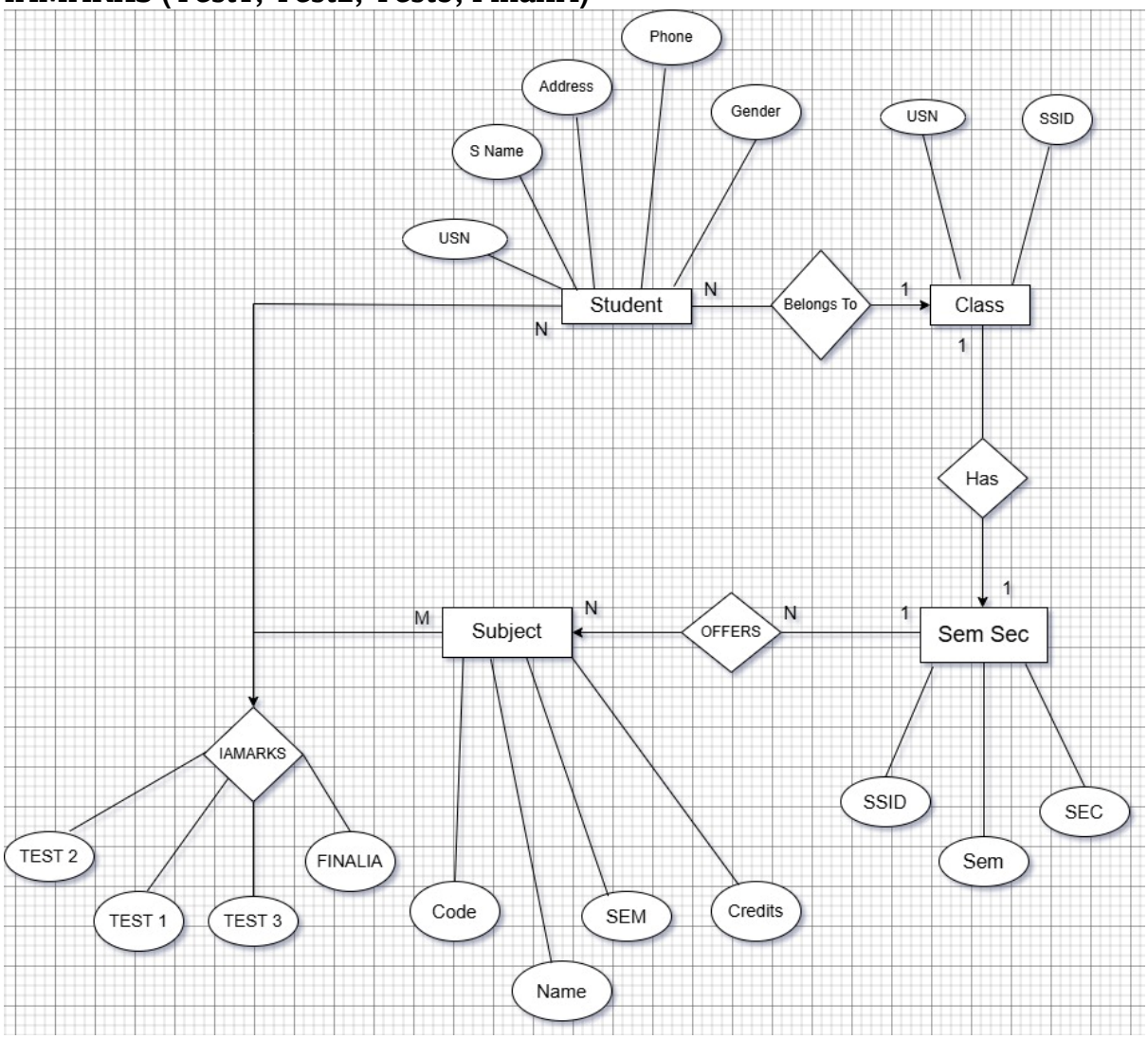
- Employees work on different projects. This many-to-many relationship is shown using the WORKS ON relationship. The number of hours an employee works on a project is stored as an attribute of this relationship.

EXPERIMENT-2

2. Consider following databases and draw ER diagram and convert entities and relationships to relation table for a given scenario:

COLLEGE DATABASE:

STUDENT (USN, S Name, Address, Phone, Gender) SEMSEC (SSID, Sem, SEC)
CLASS (USN, SSID)
SUBJECT (Code, Name, Sem, Credits)
IAMARKS (Test1, Test2, Test3, FinalIA)



Explanation of ER Diagram – College Database :

- This ER diagram represents a college database that stores information about students, classes, semester–sections, subjects, and internal

assessment marks.

- In this diagram, the Student entity stores the basic details of a student such as USN, student name, address, phone number, and gender. Every student belongs to a particular class. This relationship is shown using the BELONGS TO relationship. Many students can belong to the same class, but a student can belong to only one class.

- The Class entity represents a group of students. Each class is associated with exactly one semester and section, which is represented by the SemSec entity. The relationship between Class and SemSec is one-to-one, meaning each class has only one semester-section and each semester-section belongs to one class.

- The SemSec entity stores information about the semester and section. A semester-section offers multiple subjects. This is shown by the OFFERS relationship between SemSec and Subject. One semester-section can offer many subjects, but each subject is offered in one specific semester-section.

- The Subject entity stores subject details such as subject code, subject name, semester, and credits. Students study multiple subjects, and each subject is studied by many students. This many-to-many relationship is represented using the IAMARKS relationship.

- The IAMARKS relationship stores the internal assessment marks of students for each subject. It includes Test1, Test2, Test3, and Final IA. This relationship connects Student and Subject and stores marks for every student–subject combination.